

회귀 (Regression)

```
import numpy as np
import pandas as pd

from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error

from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.pipeline import Pipeline
```

```
# 공통 평가지표 계산
def evaluate_regression(y_true, y_pred):
    r2 = r2_score(y_true, y_pred)
    mae = mean_absolute_error(y_true, y_pred)
    rmse = np.sqrt(mean_squared_error(y_true, y_pred))
    return {"r2": r2, "mae": mae, "rmse": rmse}
```

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

```
# 1) 데이터 로드
data = load_diabetes(as_frame=True)
df = data.frame.copy()
X = df.drop(columns=["target"])
y = df["target"]
```

df

	age	sex	bmi	bp	s1	s2	s3	s4	s5	s6	target
0	0.038076	0.050680	0.061696	0.021872	-0.044223	-0.034821	-0.043401	-0.002592	0.019907	-0.017646	151.0
1	-0.001882	-0.044642	-0.051474	-0.026328	-0.008449	-0.019163	0.074412	-0.039493	-0.068332	-0.092204	75.0
2	0.085299	0.050680	0.044451	-0.005670	-0.045599	-0.034194	-0.032356	-0.002592	0.002861	-0.025930	141.0
3	-0.089063	-0.044642	-0.011595	-0.036656	0.012191	0.024991	-0.036038	0.034309	0.022688	-0.009362	206.0
4	0.005383	-0.044642	-0.036385	0.021872	0.003935	0.015596	0.008142	-0.002592	-0.031988	-0.046641	135.0
...	...	...	...	...	...	...	...	...	...	...	...
437	0.041708	0.050680	0.019662	0.059744	-0.005697	-0.002566	-0.028674	-0.002592	0.031193	0.007207	178.0
438	-0.005515	0.050680	-0.015906	-0.067642	0.049341	0.079165	-0.028674	0.034309	-0.018114	0.044485	104.0
439	0.041708	0.050680	-0.015906	0.017293	-0.037344	-0.013840	-0.024993	-0.011080	-0.046883	0.015491	132.0
440	-0.045472	-0.044642	0.039062	0.001215	0.016318	0.015283	-0.028674	0.026560	0.044529	-0.025930	220.0
441	-0.045472	-0.044642	-0.073030	-0.081413	0.083740	0.027809	0.173816	-0.039493	-0.004222	0.003064	57.0

442 rows × 11 columns

```
# 2) train / val / test 분할
X_train, X_temp, y_train, y_temp = train_test_split(
    X, y, test_size=0.4, random_state=42
)
```

```
X_val, X_test, y_val, y_test = train_test_split(
    X_temp, y_temp, test_size=0.5, random_state=42
)

print("Shapes:", X_train.shape, X_val.shape, X_test.shape)
```

Shapes: (265, 10) (88, 10) (89, 10)

X_test_1

	bmi
116	0.058463
82	-0.035307
327	0.114509
407	0.092953
388	-0.024529
...	...
415	0.008883
140	0.014272
399	0.022895
364	-0.006206
46	-0.011595

89 rows × 1 columns

A) 단순 선형회귀(단일 특성) - 컬럼 하나만 사용

```
single_feature = ["bmi"]
X_train_1 = X_train[single_feature]
X_val_1 = X_val[single_feature]
X_test_1 = X_test[single_feature]

lr_single = LinearRegression()
lr_single.fit(X_train_1, y_train)

val_pred = lr_single.predict(X_val_1)
test_pred = lr_single.predict(X_test_1)

print(evaluate_regression(y_val, val_pred))
print(evaluate_regression(y_test, test_pred))
```

```
{'r2': 0.35807192324505865, 'mae': 49.205121728237, 'rmse': np.float64(60.83306853752704)}
{'r2': 0.22294188943403737, 'mae': 55.210657137130205, 'rmse': np.float64(67.04404354626587)}
```

B) 다중회귀(여러 특성) - 모든 특성 사용

```
lr_multi = LinearRegression()
lr_multi.fit(X_train, y_train)

val_pred = lr_multi.predict(X_val)
test_pred = lr_multi.predict(X_test)

print(evaluate_regression(y_val, val_pred))
print(evaluate_regression(y_test, test_pred))
```

```
{'r2': 0.5809667034070141, 'mae': 38.22127405114377, 'rmse': np.float64(49.14969247001349)}
{'r2': 0.4389137589815666, 'mae': 46.72401655139614, 'rmse': np.float64(56.970267692676664)}
```

▽ 다항회귀 어떻게 구현하지??

$$y = w_0 + w_1x$$

$$y = w_0 + w_1x + w_2x^2$$

```
# C) 다항회귀(Polynomial Regression)
poly_degree = 2
poly_model = Pipeline([
    ("poly", PolynomialFeatures(degree=poly_degree, include_bias=False)), # 보통 회귀는 상수항 들어가서 굳이 넣을 필요 X
    ("lr", LinearRegression())
])

poly_model.fit(X_train, y_train)

val_pred = poly_model.predict(X_val)
test_pred = poly_model.predict(X_test)

print(evaluate_regression(y_val, val_pred))
print(evaluate_regression(y_test, test_pred))

{'r2': 0.41755481217394785, 'mae': 47.276965639890435, 'rmse': np.float64(57.94608094288006)}
{'r2': 0.4106135380059399, 'mae': 47.08183556189856, 'rmse': np.float64(58.389334939889366)}
```

▽ 규제 (Regularization)

$$\min_w \sum_{i=1}^n (y_i - x_i^T w)^2$$

$$\min_w \sum_{i=1}^n (y_i - x_i^T w)^2 + \lambda \sum_{j=1}^p w_j^2$$

$$\min_w \sum_{i=1}^n (y_i - x_i^T w)^2 + \lambda \sum_{j=1}^p |w_j|$$

```
from sklearn.linear_model import Ridge

# D-1) Ridge Regression
ridge = Ridge(alpha=1.0, random_state=42)
ridge.fit(X_train, y_train)

val_pred = ridge.predict(X_val)
test_pred = ridge.predict(X_test)

print(evaluate_regression(y_val, val_pred))
print(evaluate_regression(y_test, test_pred))

{'r2': 0.4431419268359479, 'mae': 47.53504404160745, 'rmse': np.float64(56.65898603535078)}
{'r2': 0.3709064412384777, 'mae': 50.127087483900425, 'rmse': np.float64(60.32413037033285)}
```

```
from sklearn.linear_model import Lasso

# D-2) Lasso Regression
lasso = Lasso(alpha=0.01, random_state=42, max_iter=10000)
lasso.fit(X_train, y_train)

val_pred = lasso.predict(X_val)
test_pred = lasso.predict(X_test)

print(evaluate_regression(y_val, val_pred))
print(evaluate_regression(y_test, test_pred))

{'r2': 0.5784115114017325, 'mae': 38.381989151071245, 'rmse': np.float64(49.29931784826742)}
{'r2': 0.4317830349897033, 'mae': 46.99650799792051, 'rmse': np.float64(57.33113618342399)}
```

▽ 다항회귀에 pipeline과 grid search까지 써보기

```
import numpy as np
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.pipeline import Pipeline
```

```

from sklearn.preprocessing import PolynomialFeatures, StandardScaler
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error

```

0) 공통 평가 함수

```

def evaluate_regression(y_true, y_pred):
    r2 = r2_score(y_true, y_pred)
    mae = mean_absolute_error(y_true, y_pred)
    rmse = np.sqrt(mean_squared_error(y_true, y_pred))
    return r2, mae, rmse

```

1) 데이터

```

data = load_diabetes(as_frame=True)
X = data.data
y = data.target

```

2) train / test 먼저 분리 (gridsearch는 train에서 CV)

```

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

```

3) 파이프라인

```

# - 다항특성 생성 -> 스케일링 -> 선형회귀
pipe = Pipeline([
    ("poly", PolynomialFeatures(include_bias=False)),
    ("scaler", StandardScaler()),
    ("lr", LinearRegression())
])

```

4) 그리드 서치 설정

```

# scoring은 R2 기준으로 최적화
param_grid = {
    "poly__degree": [1, 2, 3], # 10이면 사실상 '다중 선형회귀'
    "poly__interaction_only": [False, True]
}

```

```

grid = GridSearchCV(
    estimator=pipe,
    param_grid=param_grid,
    scoring="r2",
    cv=5,
    n_jobs=-1
)

```

```
grid.fit(X_train, y_train)
```

```

print("Best params:", grid.best_params_)
print(f"Best CV R2 : {grid.best_score_:.4f}")

```

```
best_model = grid.best_estimator_
```

5) 테스트 성능 평가

```

test_pred = best_model.predict(X_test)
r2, mae, rmse = evaluate_regression(y_test, test_pred)

```

```

print("\n[Best Model - TEST]")
print(f"R2 : {r2:.4f}")
print(f"MAE : {mae:.4f}")
print(f"RMSE : {rmse:.4f}")

```

```

Best params: {'poly__degree': 1, 'poly__interaction_only': False}
Best CV R2 : 0.4493

```

```

[Best Model - TEST]
R2 : 0.4526
MAE : 42.7941
RMSE : 53.8534

```

▽ 위 과정을 경사하강법으로 진행?

```
import numpy as np
```

```

from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.pipeline import Pipeline

```

```

from sklearn.preprocessing import PolynomialFeatures, StandardScaler
from sklearn.linear_model import SGDRegressor
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error

```

0) 평가 함수

```

def evaluate_regression(y_true, y_pred):
    r2 = r2_score(y_true, y_pred)
    mae = mean_absolute_error(y_true, y_pred)
    rmse = np.sqrt(mean_squared_error(y_true, y_pred))
    return r2, mae, rmse

```

1) 데이터

```

data = load_diabetes(as_frame=True)
X = data.data
y = data.target

```

2) train / test 분리

```

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

```

3) 파이프라인

```

pipe = Pipeline([
    ("poly", PolynomialFeatures(include_bias=False)),
    ("scaler", StandardScaler()),
    ("sgd", SGDRegressor(random_state=42))
])

```

4) 그리드 설정

```

param_grid = {
    "poly__degree": [1, 2, 3],
    "poly__interaction_only": [False, True],

    "sgd__penalty": ["l2", "l1", "elasticnet"],
    "sgd__alpha": [0.0001, 0.001, 0.01],
    "sgd__max_iter": [1000, 2000],
    "sgd__learning_rate": ["constant", "optimal"],
    "sgd__eta0": [0.001, 0.01] # constant일 때 사용
}

```

```

grid = GridSearchCV(
    estimator=pipe,
    param_grid=param_grid,
    scoring="r2",
    cv=5,
    n_jobs=-1
)

```

```
grid.fit(X_train, y_train)
```

```

print("Best params:", grid.best_params_)
print(f"Best CV R2 : {grid.best_score_:.4f}")

```

```
best_model = grid.best_estimator_
```

5) 테스트 성능 평가

```
test_pred = best_model.predict(X_test)
```

```
r2, mae, rmse = evaluate_regression(y_test, test_pred)
```

```

print("\n[Best Model - TEST]")
print(f"R2 : {r2:.4f}")
print(f"MAE : {mae:.4f}")
print(f"RMSE : {rmse:.4f}")

```

```

Best params: {'poly__degree': 1, 'poly__interaction_only': False, 'sgd__alpha': 0.01, 'sgd__eta0': 0.001, 'sgd__learning_rate': 'constant'}
Best CV R2 : 0.4522

```

```

[Best Model - TEST]
R2 : 0.4559
MAE : 42.9018
RMSE : 53.6925

```

▽ 주요 하이퍼파라미터

1. loss

- 기본값: 'squared_error'
- squared_error → 일반 선형회귀

- huber → 이상치에 강함
2. penalty
- 규제 방식
 - 'l2' → Ridge
 - 'l1' → Lasso
 - 'elasticnet' → 둘 다
 - None → 정규화 없음
3. alpha
- 규제 강도 (λ)
 - 값이 클수록:
 - 가중치 작아짐, 과적합 감소, bias 증가
4. learning_rate
- 학습률 전략
 - 'constant'
 - 항상 같은 학습률 사용
 - 'optimal' (기본값)
 - 점점 감소하는 학습률
 - 이론적으로 수렴 보장, 자동으로 적절히 감소, 안정적
 - 'invscaling'
 - 점진적 감소
 - optimal보다 감소 속도 조절 가능(power_t의 기본값 0.25인데 이걸로 조절 가능)
 - 'adaptive'
 - 초기에는 constant처럼 유지
 - 검증 결과가 개선되지 않으면 학습률을 줄임
5. eta0
- 초기 학습률
 - learning_rate='constant' 일 때 중요
6. max_iter
- 최대 반복 횟수
 - SGD는 반복 학습이라 매우 중요
7. tol
- 수렴 조건
 - 변화량이 tol보다 작으면 중단
8. early_stopping
- True로 하면:
 - 내부 validation split
 - 성능 나빠지면 자동 종료
 - 딥러닝 느낌의 안정성 옵션

코딩을 시작하거나 AI로 코드를 생성하세요.

